SST 409: Stochastic Processes

Complete Lecture Notes — Random Variables and Generating Functions

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*A merged transcription of two handwritten note sets for SST 409: the seven-page "full notes" set (referred to below as the **extended set**) and the fifteen-page "lecture notes on random variables" set (the **lecture notes**). Shared material has been merged into one continuous treatment using the fuller of the two versions; material found only in the lecture notes (sums with a random number of terms, tail generating functions, bivariate generating functions) forms Part II. All equations, symbols and notation are typeset in LaTeX, and Appendix B maps every source page to the section it feeds.*

Abbreviations as used in the originals: r.v. = random variable; st. pr. = stochastic process; g.f. = generating function; p.g.f. = probability generating function; dist./distr. = distribution; m.g.f. = moment generating function; mtd. = method; qty = quantity; prob. = probability; nos = numbers.

— extended set, p. 1 —

Part I. Foundations and Generating Functions

1. Stochastic Processes

1.1 Definition

- By stochastic processes, we mean a set of r.v. $\{X(t)\}$ depending on a real parameter t which varies in a certain set of natural nos.
- A stochastic process is denoted by

$$\{X(t),\ t \in T\}.$$

In many problems of theory of st. pr., the parameter t means time. Though it can denote any other physical qty such as space in which the process occurs.

1.2 Examples of stochastic processes

i) Random walk. Where we are concerned with the motion of a particle in a straight line which moves in discrete jumps with certain probabilities i.e. at each stage, it moves one unit right to left with probability p or $1 - p$.

If initially the particle is at the origin, then its position $X(n)$ after n jumps is a discrete r.v. depending on the discrete parameter n. Thus $\{X(n),\ t \in T\}$ is a stochastic process.

ii) Number of telephone calls. The no. of telephone calls $X(t)$ during the arbitrary period of the time t is a discrete r.v. depending on the continuous parameter t. Thus $\{X(t),\ t \in T\}$ is a stochastic process.

(Examples iii and iv appear only in the extended set.)

iii) Water level of a river. The water level of a river under a certain bridge measured daily at any moment t is a continuous r.v. depending on the continuous parameter t. Thus $\{X(t),\ t \in T\}$ is a stochastic process.

iv) Brownian motion. The no. of particles executing a Brownian motion with small, rapid random steps is a discontinuous r.v. depending on the continuous parameter t. Thus $\{X(t),\ t \in T\}$ constitutes a stochastic process.

2. Generating Functions

2.1 Definition

Let $a_0, a_1, a_2, \ldots$ be a sequence of real numbers. Using a variable s we may define a function

$$A(s) = a_0 + a_1 s + a_2 s^2 + \cdots = \sum_{k=0}^{\infty} a_k s^k \qquad \cdots(1)$$

If the power series $A(s)$ converges in some interval $s_0 < s < s_1$ (the extended set writes $-s_0 < s < s_0$), then $A(s)$ is called the generating function (g.f.) of the sequence $a_0, a_1, a_2, \ldots$. The variable s itself has no particular significance.

2.2 Extracting the coefficients by differentiation

Now:

$$\frac{dA(s)}{ds} = a_1 + 2a_2 s + 3a_3 s^2 + 4a_4 s^3 + \cdots + k a_k s^{k-1} + \cdots$$

$$\left.\frac{dA(s)}{ds}\right|_{s=0} = a_1$$

$$\frac{d^2A(s)}{ds^2} = 2a_2 + 3\times 2\, a_3 s + 4\times 3\, a_4 s^2 + \cdots + k(k-1) a_k s^{k-2} + \cdots$$

$$\left.\frac{d^2A(s)}{ds^2}\right|_{s=0} = 2a_2 \quad \Rightarrow \quad a_2 = \frac{1}{2}\left.\frac{d^2A(s)}{ds^2}\right|_{s=0}$$

$$\left.\frac{d^3A(s)}{ds^3}\right|_{s=0} = 3!\, a_3 \quad \Rightarrow \quad a_3 = \frac{1}{3!}\left.\frac{d^3A(s)}{ds^3}\right|_{s=0}$$

$$\left.\frac{d^kA(s)}{ds^k}\right|_{s=0} = k!\, a_k \quad \Rightarrow \quad a_k = \frac{1}{k!}\left.\frac{d^kA(s)}{ds^k}\right|_{s=0}$$

2.3 Examples

Exp 1. If $a_k = 1$ for all k, i.e. $\{a_k\} = \{1, 1, 1, \ldots\}$, then $a_0 = 1, a_1 = 1, a_2 = 1, \ldots$

The generating function of $\{a_k\}$ is

$$A(s) = a_0 + a_1 s + a_2 s^2 + \cdots = 1 + s + s^2 + \cdots = \frac{1}{1-s}, \qquad |s| < 1$$

Exp 2. If $\{a_k\} = \left\{\frac{1}{k!}\right\}$, then $a_0 = 1, a_1 = 1, a_2 = \frac{1}{2!}, a_3 = \frac{1}{3!}, \ldots$

The g.f. of $\{a_k\}$ is

$$A(s) = a_0 + a_1 s + a_2 s^2 + \cdots = \frac{1}{0!} + \frac{1}{1!}s + \frac{1}{2!}s^2 + \cdots = 1 + s + \frac{s^2}{2!} + \frac{s^3}{3!} + \cdots = e^s$$

Exp 3. Let $\{a_k\} = \left\{\binom{n}{k}\right\}$ for fixed n.

The g.f. of $\{a_k\}$ is

$$A(s) = \sum_{k=0}^{\infty} a_k s^k = \sum_{k=0}^{\infty} \binom{n}{k} s^k = \sum_{k=0}^{n} \binom{n}{k} s^k \qquad (\text{any value} > n \text{ is zero}) = (1+s)^n$$

Exp 4. Let $a_0, a_1, a_2, \ldots, a_n$ be a sequence of numbers satisfying the recurrence relation

$$a_n + a_{n-1} - 16a_{n-2} + 20a_{n-3} = 0 \qquad \text{for } n \geq 3$$

subject to the initial values $a_0 = 0, a_1 = 1$ & $a_2 = -1$. Obtain the generating function of the above sequence $\{a_n\}$.

Soln

Let $A(s) = \sum_{n=0}^{\infty} a_n s^n$ be the g.f. of $\{a_n\}$.

$$\sum_{n=3}^{\infty} a_n s^n + \sum_{n=3}^{\infty} a_{n-1} s^n - 16 \sum_{n=3}^{\infty} a_{n-2} s^n + 20 \sum_{n=3}^{\infty} a_{n-3} s^n = 0$$

$$\left(A(s) - a_0 - a_1 s - a_2 s^2\right) + s \sum_{n=3}^{\infty} a_{n-1} s^{n-1} - 16s^2 \sum_{n=3}^{\infty} a_{n-2} s^{n-2} + 20s^3 \sum_{n=3}^{\infty} a_{n-3} s^{n-3} = 0$$

$$\left(A(s) - s + s^2\right) + s\left(A(s) - a_0 - a_1 s\right) - 16s^2\left(A(s) - a_0\right) + 20s^3 A(s) = 0$$

$$A(s) - s + s^2 + sA(s) - s^2 - 16s^2 A(s) + 20s^3 A(s) = 0$$

$$A(s)\left(1 + s - 16s^2 + 20s^3\right) = s$$

$$\therefore \quad A(s) = \frac{s}{1 + s - 16s^2 + 20s^3} \qquad \text{is the g.f. of } \{a_n\}$$

— extended set, p. 3 —

3. Probability Generating Function

3.1 Definition

Suppose that X is a r.v. which assumes non-negative integral values 0, 1, 2, … & that $\Pr(X = k) = p_k, k = 0, 1, 2, \ldots, \sum p_k = 1$; then the probability generating function $P(s)$ of the sequence of probabilities $\{p_k\}$ is given by:

$$P(s) = \sum_{k=0}^{\infty} p_k s^k = p_0 + p_1 s + p_2 s^2 + \cdots + p_k s^k + \cdots$$

$$= \sum_{k=0}^{\infty} \Pr(X = k)\, s^k = E\left(s^X\right) = F\left(s^x\right)$$

where $E(s^X)$ — written $F(s^x)$ in the lecture notes — is the expectation of the function s^X (a r.v.) of the r.v. X.

The series $P(s)$ converges for at least $-1 \leq s \leq 1$. Clearly $P(1) = 1$.

3.2 A sequence that is not a p.g.f.

NB: It is possible to have a sequence of probabilities which does not form a p.g.f.

e.g. If $p_0 = 0, p_1 = \frac{1}{4}, p_2 = \frac{1}{2}, p_3 = \frac{1}{3}, p_4 = \frac{3}{4}$

$$A(s) = p_0 + p_1 s + p_2 s^2 + p_3 s^3 + p_4 s^4 = \frac{1}{4}s + \frac{1}{2}s^2 + \frac{1}{3}s^3 + \frac{3}{4}s^4$$

$A(s)$ is a g.f. of $\{p_k\}$ but not a p.g.f., since

$$p_0 + p_1 + p_2 + p_3 + p_4 \neq 1.$$

3.3 Determining mean & variance using the p.g.f.

$$P(s) = \sum_{k=0}^{\infty} p_k s^k$$

$$P'(s) = \sum_k k\,p_k s^{k-1}, \qquad -1 < s < 1$$

$$\& \qquad P''(s) = \sum_k k(k-1)\,p_k s^{k-2}$$

The Expectation $E(x)$ is given by

$$E(x) = P'(1) = \sum_k k\,p_k$$

$$\& \qquad E\left(x^2\right) = E\left[x(x-1)+x\right] = E\left[x(x-1)\right] + E(x)$$

But

$$E\left[x(x-1)\right] = P''(1) = \sum_k k(k-1)\,p_k$$

$$E\left(x^2\right) = P''(1) + P'(1)$$

Hence

$$\mathrm{Var}(x) = E\left(x^2\right) - \left[E(x)\right]^2 = P''(1) + P'(1) - \left[P'(1)\right]^2$$

4. The p.g.f. & Moments of Standard Distributions

4.1 Bernoulli Distribution

Let x have a Bernoulli distr. with parameter p i.e.

$$\Pr\{X = k\} = p_k = p^k q^{1-k}, \qquad q = 1-p, \qquad k = 0, 1.$$

The p.g.f. of x is given by

$$P(s) = \sum_k p_k s^k = \sum_{k=0}^{1} p^k q^{1-k} s^k = q + ps$$

$$P'(s) = p \qquad P''(s) = 0$$

$$E(x) = P'(1) = p \qquad P''(1) = 0$$

$$\mathrm{Var}(x) = P''(1) + P'(1) - \left[P'(1)\right]^2 = 0 + p - p^2 = p(1-p) = pq$$

since $p + q = 1$.

4.2 Binomial Distribution

$$\Pr\{X = k\} = p_k = \binom{n}{k} p^k q^{n-k}, \qquad k = 0, 1, \ldots, n, \qquad q + p = 1.$$

The p.g.f. of x is given by

$$P(s) = \sum_k p_k s^k = \sum_{k=0}^{n} \binom{n}{k} p^k q^{n-k} s^k = (q + ps)^n$$

$$P'(s) = n(q+ps)^{n-1} p = np\,(q+ps)^{n-1}$$

$$E(x) = P'(1) = np\,(q+p)^{n-1} = np \qquad \text{since } q + p = 1.$$

— extended set, p. 5 —

$$P''(s) = np(n-1)(q+ps)^{n-2} p$$

$$P''(1) = n(n-1)p^2(q+p)^{n-2} = n(n-1)p^2 \qquad \text{since } q + p = 1.$$

$$\mathrm{Var}(x) = P''(1) + P'(1) - \left[P'(1)\right]^2$$

$$= n(n-1)p^2 + np - (np)^2$$
$$= n^2p^2 - np^2 + np - n^2p^2 = np(1-p) = npq$$

4.3 Geometric Distribution

$$\Pr\{X = k\} = p_k = q^k p, \qquad p + q = 1, \qquad k = 0, 1, 2, \ldots$$

The p.g.f. of x is given by

$$P(s) = \sum_{k=0}^{\infty} p_k s^k = \sum_{k=0}^{\infty} q^k p\, s^k = p \sum_{k=0}^{\infty} (qs)^k$$
$$= p\left(1 + qs + (qs)^2 + (qs)^3 + \cdots\right) = p\left(\frac{1}{1-qs}\right) = \frac{p}{1-qs}$$
$$P'(s) = \frac{pq}{(1-qs)^2} \qquad P''(s) = \frac{2pq^2}{(1-qs)^3}$$
$$E(x) = P'(1) = \frac{pq}{(1-q)^2} = \frac{pq}{p^2} = \frac{q}{p} \qquad P''(1) = \frac{2pq^2}{(1-q)^3} = \frac{2q^2}{(1-q)^3} = \frac{2pq^2}{p^3}$$
$$\mathrm{Var}(x) = P''(1) + P'(1) - \left[P'(1)\right]^2$$
$$= \frac{2q^2}{p^2} + \frac{q}{p} - \frac{q^2}{p^2}$$
$$= \frac{q^2}{p^2} + \frac{q}{p} = \frac{q^2 + pq}{p^2} = \frac{q(q+p)}{p^2} = \frac{q}{p^2}, \qquad (p+q) = 1.$$

4.4 Poisson Distribution

(The lecture notes mark the Poisson case "Read"; the derivation below is from the extended set.)

$$\Pr(X = k) = p_k = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

The p.g.f. of x is given by

$$\sum_{k=0}^{\infty} p_k s^k = \sum_{k=0}^{\infty} e^{-\lambda}\frac{\lambda^k}{k!} s^k$$
$$P(s) = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda s)^k}{k!} = e^{-\lambda}\left[1 + \lambda s + \frac{(\lambda s)^2}{2!} + \cdots\right] = e^{-\lambda} e^{\lambda s} = e^{-\lambda(1-s)}$$
$$P'(s) = \lambda e^{-\lambda(1-s)} \qquad P''(s) = \lambda^2 e^{-\lambda(1-s)}$$
$$E(x) = P'(1) = \lambda e^{-\lambda(1-1)} = \lambda e^0 = \lambda \qquad P''(1) = \lambda^2 e^{-\lambda(1-1)} = \lambda^2 e^0 = \lambda^2$$
$$\mathrm{Var}(x) = P''(1) + P'(1) - \left[P'(1)\right]^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

— extended set, p. 6 —

5. Remarks on the p.g.f.

5.1 Link with the moment generating function

i) When $s = e^t$ then

$$P(s) = P\left(e^t\right) = E\left(e^{tx}\right) \qquad \text{which is the m.g.f. of } x.$$

5.2 The integral of the p.g.f.

ii)

$$\int_0^1 P(s)\,ds = \int_0^1 \left(\sum_{k=0}^{\infty} p_k s^k \right) ds$$

$$= \int_0^1 \left(p_0 + p_1 s + p_2 s^2 + p_3 s^3 + p_4 s^4 + \cdots \right) ds$$

$$= \int_0^1 p_0\,ds + \int_0^1 p_1 s\,ds + \int_0^1 p_2 s^2\,ds + \int_0^1 p_3 s^3\,ds + \int_0^1 p_4 s^4\,ds + \cdots$$

$$= p_0 + \frac{p_1}{2} + \frac{p_2}{3} + \frac{p_3}{4} + \frac{p_4}{5} + \frac{p_5}{6} + \cdots$$

$$= \sum_{k=0}^{\infty} \frac{p_k}{k+1} = E\left(\frac{1}{X+1} \right)$$

5.3 Example: the p.g.f. of $Y = mX + n$

Let X be a r.v. with p.g.f. $P(s)$. Find the p.g.f. of $Y = mX + n$, where m, n are integers & $m \neq 0$.

Let $P_Y(s)$ be the p.g.f. of Y. Then

$$P_Y(s) = E\left(s^Y \right) = E\left[s^{mX+n} \right]$$

$$= E\left[s^{mX} s^n \right]$$

$$= s^n E\left[\left(s^m \right)^X \right] = s^n P\left(s^m \right)$$

(with $P_x(s) = E(s^x)$ as noted in the margin.)

— extended set, p. 7 —

5.4 Recovering the probabilities from the p.g.f.

Remark i) We can obtain probabilities p_k from the p.g.f. p_k can be found from $P(s)$ i.e.

$$p_k = \frac{1}{k!} \left[\frac{d^k P(s)}{ds^k} \bigg|_{s=0} \right]$$

Remark ii) p_k is also given by the coefficient of s^k in the expansion of $P(s)$ as a power series e.g.

If

$$P(s) = e^{-\lambda(1-s)} = e^{-\lambda} e^{\lambda s} = e^{-\lambda} \left[1 + \frac{\lambda s}{1!} + \frac{(\lambda s)^2}{2!} + \cdots + \frac{(\lambda s)^k}{k!} + \cdots \right]$$

then

$$p_k = \text{coefficient of } s^k = e^{-\lambda} \frac{\lambda^k}{k!}$$

which is a poisson distr. with parameter λ.

6. Sum of Random Variables: (i) Fixed Number

Let X & Y be two independent r.v.'s (non-negative integral r.v.'s) with prob. distributions

$$a_k = P(X = k) \qquad \& \qquad b_j = P(Y = j)$$

Let $P_x(s)$ be the p.g.f. of X & let $P_y(s)$ be the p.g.f. of Y. If the p.g.f. of $Z = x + y$ is $P_z(s)$,

$$P_z(s) = E\left(s^Z\right) = E\left(s^{x+y}\right)$$
$$= E\left(s^x s^y\right) = E\left(s^x\right) E\left(s^y\right) \qquad \text{since } X \ \& Y \text{ are independent}$$
$$= P_x(s)\, P_y(s)$$

This result says that the p.g.f. of the sum of 2 independent r.v. X & Y is the product of the p.g.f. of X & that of Y.

The result extends to the case when we have n independent r.v.'s $x_1, x_2, \ldots, x_n$, i.e. the p.g.f. of $Z = x_1 + x_2 + \cdots + x_n$ is the product of the p.g.f.'s of the individual r.v.'s:

$$P_{S_n}(s) = E\left[s^{S_n}\right] = E\left[s^{x_1 + x_2 + \cdots + x_n}\right]$$
$$= E\left(s^{x_1}\right) E\left(s^{x_2}\right) \cdots E\left(s^{x_n}\right) \qquad \text{since } x_1, \ldots, x_n \text{ are independent}$$
$$= P_{x_1}(s)\, P_{x_2}(s) \cdots P_{x_n}(s)$$

— lecture notes, p. 9 —

If $x_1, x_2, \ldots, x_n$ are also identically distributed, each with p.g.f. $P(s)$, then the p.g.f. of $S_n = x_1 + x_2 + \cdots + x_n$ is

$$\left[P(s)\right]^n$$

Exp: the sum of two Poisson variables

Let x_1 & x_2 be indep. poisson r.v.'s with parameters λ_1 & λ_2 resp. Find the distribution of $Z = X_1 + X_2$.

Soln

Let $P_{x_1}(s)$ & $P_{x_2}(s)$ be the p.g.f.'s of X_1 & X_2 respectively. Then

$$P_{x_1}(s) = \sum_{k=0}^{\infty} p_k s^k = \sum_{k=0}^{\infty} e^{-\lambda_1} \frac{\lambda_1^k}{k!} s^k = e^{-\lambda_1} \sum_{k=0}^{\infty} \frac{(\lambda_1 s)^k}{k!} = e^{-\lambda_1} e^{\lambda_1 s} = e^{-\lambda_1(1-s)}$$

and similarly $P_{x_2}(s) = e^{-\lambda_2(1-s)}$.

∴ The p.g.f. of Z is

$$P_z(s) = P_{x_1}(s)\, P_{x_2}(s) = e^{-(\lambda_1 + \lambda_2)(1-s)}$$

This is the p.g.f. of a poisson r.v. with parameters λ_1 & λ_2. This shows the sum of the two indep. poisson r.v.'s is also a poisson r.v.

This result extends to the case of n indep. poisson r.v.'s. That is the sum $S_n = x_1 + x_2 + \cdots + x_n$ of n indep. poisson r.v.'s is a poisson r.v. with mean

$$\sum_{i=1}^{n} \lambda_i$$

where λ_i is the mean of x_i.

Part II. Further Topics (Lecture Notes)

7. Sum of Random Variables: (ii) Random Number

Theorem: Let x_i, $i = 1, 2, \ldots$ be identically & independently dist. (i.i.d.) random variables with $P(x_i = k) = p_k$ & p.g.f.

$$P(s) = \sum_{k=0}^{\infty} p_k s^k, \qquad i = 1, 2, 3, \ldots$$

Let $S_N = x_1 + x_2 + \cdots + x_N$ where N is a r.v. independent of the x_i's. Let the distribution of N be given by $P[N = n] = g_n$ & p.g.f.

— lecture notes, p. 10 —

of N be

$$G(s) = \sum_{n=0}^{\infty} g_n s^n.$$

Then the p.g.f., $H(s)$, of S_N is given by the compound function $G\big[P(s)\big]$ i.e.

$$H(s) = \sum_{j=0}^{\infty} P(S_N = j)\, s^j = G\big[P(s)\big]$$

$$H'(s) = G'\big[P(s)\big]P'(s)$$

$$\therefore \quad E(S_N) = H'(1) = G'\big[P(1)\big] \cdot P'(1) = G'(1)\, P'(1) = E(N)\, E(x_i)$$

Exp: Poisson number of terms

Suppose N has a poisson dist. with mean λ & x_i, $i = 1, 2, \ldots, N$ are i.i.d. with p.g.f. $P(s)$. Find the p.g.f. of $S_N = x_1 + x_2 + \cdots + x_N$.

Soln

p.g.f. of S_N is

$$H(s) = G\big[P(s)\big]$$

But p.g.f. of a poisson r.v. with mean λ is given by

$$G(s) = e^{-\lambda(1-s)}$$

$$\therefore \quad H(s) = e^{-\lambda\left(1 - P(s)\right)}$$

The mean is given by

$$E(S_N) = E(x_i)\, E(N) = \lambda E(x)$$

8. Tailed or Cumulative Probabilities

8.1 Upper tail

Let X have a p.d.f. $\Pr\{X = k\} = p_k$, $k = 0, 1, 2, \ldots$ with p.g.f.

$$P(s) = \sum_{k=0}^{\infty} p_k s^k,$$

and

$$q_k = \Pr\{X > k\} = p_{k+1} + p_{k+2} + \cdots, \qquad k = 0, 1, 2, \ldots$$

with generating funct.

$$\phi(s) = \sum_{k=0}^{\infty} q_k s^k \qquad \left(\sum_k q_k = 1 \right)$$

$$\therefore \quad \phi(s) = \sum_{k=0}^{\infty} q_k s^k = q_0 + q_1 s + q_2 s^2 + q_3 s^3 + \cdots$$

$$= (p_1 + p_2 + p_3 + \cdots) + (p_2 + p_3 + p_4 + \cdots)s + (p_3 + p_4 + p_5 + \cdots)s^2 + (p_4 + p_5 + p_6 + \cdots)s^3 + \cdots$$

— lecture notes, p. 11 —

$$= p_1 + p_2(1+s) + p_3\left(1 + s + s^2\right) + p_4\left(1 + s + s^2 + s^3\right) + \cdots$$

$$= p_1 + p_2(1+s) + p_3\frac{\left(1 - s^3\right)}{(1-s)} + p_4\frac{\left(1 - s^4\right)}{(1-s)} + \cdots$$

$$= \frac{p_1(1-s) + p_2\left(1 - s^2\right) + p_3\left(1 - s^3\right) + p_4\left(1 - s^4\right) + \cdots}{(1-s)}$$

$$= \frac{(p_1 + p_2 + p_3 + p_4 + \cdots) - \left(p_1 s + p_2 s^2 + p_3 s^3 + p_4 s^4 + \cdots\right)}{(1-s)}$$

$$= \frac{(p_0 + p_1 + p_2 + p_3 + p_4 + \cdots) - \left(p_0 + p_1 s + p_2 s^2 + p_3 s^3 + p_4 s^4 + \cdots\right)}{(1-s)}$$

$$\therefore \quad \phi(s) = \frac{1 - P(s)}{1 - s}, \qquad -1 < s < 1$$

Alt. mtd.

$$q_k = p_{k+1} + p_{k+2} + p_{k+3} + \cdots$$
$$q_{k-1} = p_k + p_{k+1} + p_{k+2} + p_{k+3} + \cdots$$
$$p_k = q_{k-1} - q_k, \qquad k \geq 1$$

Defining $P(s) = \sum_{k=0}^{\infty} p_k s^k$ & $\phi(s) = \sum_{k=0}^{\infty} q_k s^k$

$$\Rightarrow \quad \sum_{k=1}^{\infty} p_k s^k = \sum_{k=1}^{\infty} q_{k-1} s^k - \sum_{k=1}^{\infty} q_k s^k$$

$$P(s) - p_0 = s\sum_{k=1}^{\infty} q_{k-1} s^{k-1} - \left[\phi(s) - q_0\right]$$

$$P(s) - p_0 = s\,\phi(s) - \phi(s) + q_0 \qquad q_0 = p_1 + p_2 + p_3 + \cdots$$

$$P(s) - (p_0 + q_0) = \phi(s)(s-1)$$

$$\phi(s)(1-s) = (p_0 + q_0) - P(s)$$

$$\phi(s)(1-s) = 1 - P(s) \qquad \text{since } p_0 + q_0 = 1$$

$$\therefore \quad \phi(s) = \frac{1 - P(s)}{1 - s}, \qquad -1 < s < 1$$

8.2 Lower tail

— lecture notes, p. 12 —

Let $q_k = \Pr\{X \leq k\}, k = 0, 1, 2, \ldots$ & $p_k = \Pr\{X = k\}, k = 0, 1, 2, \ldots$

Defining $P(s) = \sum_{k=0}^{\infty} p_k s^k$ & $\phi(s) = \sum_{k=0}^{\infty} q_k s^k$

$$\therefore \quad q_k = p_k + p_{k-1} + p_{k-2} + \cdots + p_1 + p_0$$
$$q_{k-1} = p_{k-1} + p_{k-2} + \cdots + p_1 + p_0$$
$$\therefore \quad q_k - q_{k-1} = p_k, \qquad k \geq 1 \qquad \Rightarrow (1)$$

Multiplying both sides of the eqn (1) above by s^k & summing over k we obtain;

$$\sum_{k=1}^{\infty} q_k s^k - \sum_{k=1}^{\infty} q_{k-1} s^k = \sum_{k=1}^{\infty} p_k s^k$$

$$\therefore \quad \phi(s) - q_0 - s\,\phi(s) = P(s) - p_0$$

$$\phi(s)(1-s) = P(s) - p_0 + q_0$$

$$\phi(s)(1-s) = P(s) \qquad \text{since } p_0 = q_0$$

$$\therefore \quad \phi(s) = \frac{P(s)}{1-s}, \qquad -1 < s < 1$$

9. Bivariate Probability Generating Functions

Suppose X & Y are integral valued r.v.'s with joint prob. dist.

$$\Pr\{X = j, Y = k\} = p_{jk}, \qquad j = 0, 1, 2, \ldots; \quad k = 0, 1, 2, \ldots$$

and

$$\sum_j \sum_k p_{jk} = 1.$$

Then the joint p.g.f. of X & Y is given by

$$P(s_1, s_2) = \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} p_{jk}\, s_1^j s_2^k \qquad \text{assuming convergence}$$

for the values of s_1 & s_2 i.e. $|s_1| < 1$, $|s_2| < 1$.

$$P(s_1, s_2) = E\left(s_1^X s_2^Y\right) \quad \Rightarrow (1)$$

If $s_1 = s_2 = s$, then (1) becomes

$$P(s, s) = E\left(s^X s^Y\right) = E\left(s^{X+Y}\right)$$

If X & Y are mutually indep. r.v.'s then $P(s_1, s_2)$ becomes

$$P(s_1, s_2) = E\left(s_1^X\right) E\left(s_2^Y\right)$$

If X & Y are i.i.d., with a common p.g.f. $G(s)$, then we have

$$P(s_1, s_2) = E\left(s_1^X\right) E\left(s_2^Y\right) = G(s_1) \cdot G(s_2) = \left[G(s)\right]^2$$

9.1 P.g.f.'s of marginal distributions

Let X & Y be 2 r.v.'s, then

$$P(s_1, s_2) = E\left(s_1^X s_2^Y\right) \quad \rightarrow \text{joint p.g.f.}$$

The p.g.f. of X is given by

$$P(s_1, 1) = E\left(s_1^X\right)$$

Similarly, the p.g.f. of Y is given by

$$P(1, s_2) = E\left(s_2^Y\right)$$

— lecture notes, p. 14 —

9.2 Means, variances & co-variances from the joint p.g.f.

$P(s_1, s_2)$ → joint p.g.f. (x &y)

$P(s_1, 1) = P(s_1)$ — p.g.f. of x

$P(1, s_2) = P(s_2)$ — p.g.f. of y

$$\therefore \quad E(x) = P'(s_1)\big|_{s_1 = 1} = \left.\frac{\partial P(s_1, 1)}{\partial s_1}\right|_{s_1 = 1}$$

$$E(Y) = P'(s_2)\big|_{s_2 = 1} = \left.\frac{\partial P(1, s_2)}{\partial s_2}\right|_{s_2 = 1}$$

The second moment is:

$$E\big[X(X-1)\big] = \left.\frac{\partial^2 P(s_1, 1)}{\partial s_1^2}\right|_{s_1 = 1}$$

and

$$E\big[Y(Y-1)\big] = \left.\frac{\partial^2 P(1, s_2)}{\partial s_2^2}\right|_{s_2 = 1}$$

$$\sigma_x^2 = E\big[X(X-1)\big] + E(x)^2 - \big[E(x)\big]^2$$

$$\sigma_y^2 = E\big[Y(Y-1)\big] + E(Y) - \big[E(Y)\big]^2$$

$$\operatorname{cov}(x, y) = \sigma_{xy} = E(xy) - E(x)E(y)$$

but

$$E(xy) = \left.\frac{\partial^2 P(s_1, s_2)}{\partial s_1 \partial s_2}\right|_{s_1 = s_2 = 1}$$

9.3 Exp: failures before and between successes

Consider a series of Bernoulli trials with prob. of success p. Suppose that X denotes the no. of failures preceeding the 1st success & Y the no. of failures following 1st success & preceeding 2nd success. Then the joint prob. distribution of X & Y is given by

$$p_{jk} = \Pr\{X = j, Y = k\} = q^{j+k}p^2; \qquad j, k = 0, 1, 2; \qquad p + q = 1$$

— lecture notes, p. 15 —

Soln

The joint p.g.f. is

$$P(s_1, s_2) = \sum_{j=0}^{\infty}\sum_{k=0}^{\infty} p_{jk}\, s_1^j s_2^k = \sum_{j=0}^{\infty}\sum_{k=0}^{\infty} q^{j+k}p^2\, s_1^j s_2^k$$

$$= p^2 \sum_{j=0}^{\infty} (qs_1)^j \sum_{k=0}^{\infty} (qs_2)^k$$

$$= p^2\Big[\Big(1 + qs_1 + (qs_1)^2 + \cdots\Big)\Big(1 + qs_2 + (qs_2)^2 + \cdots\Big)\Big]$$

$$P(s_1, s_2) = p^2\left(\frac{1}{1 - qs_1}\right)\left(\frac{1}{1 - qs_2}\right) = \frac{p^2}{(1 - qs_1)(1 - qs_2)}$$

$$P(s_1) = P(s_1, 1) = \frac{p^2}{(1 - qs_1)(1 - q)} = \frac{p}{1 - qs_1}, \qquad \text{since } q + p = 1$$

$$\therefore \quad P'(s_1) = \frac{pq}{(1 - qs_1)^2}$$

$$\therefore \quad E(x) = P'(1) = \frac{pq}{(1 - q)^2} = \frac{pq}{p^2} = \frac{q}{p}$$

$$P''(s_1) = \frac{2pq^2}{(1-qs_1)^3} \qquad P''(1) = \frac{2pq^2}{(1-q)^3} = \frac{2pq^2}{p^3} = \frac{2q^2}{p^2}$$

$$\therefore \quad \text{Var}(x) = P''(1) + P'(1) - \left[P'(1)\right]^2$$

$$= \frac{2q^2}{p^2} + \frac{q}{p} - \frac{q^2}{p^2} = \frac{q^2}{p^2} + \frac{q}{p}$$

$$= \frac{q^2+pq}{p^2} = \frac{q(q+p)}{p^2} = \frac{q}{p^2} \qquad \text{since } q+p=1$$

Similarly

$$E(y) = \frac{q}{p} \qquad \& \qquad \text{Var}(y) = \frac{q}{p^2}$$

$$\frac{\partial P(s_1,s_2)}{\partial s_1} = \frac{p^2 q}{(1-qs_1)^2(1-qs_2)}$$

$$\therefore \quad \frac{\partial^2 P(s_1,s_2)}{\partial s_1 \partial s_2} = \frac{p^2q^2}{(1-qs_1)^2(1-qs_2)^2}$$

Appendix A. Notes on the Originals

Minor notational slips in the handwritten originals. They are transcribed above as written; the corrected forms are given here for study purposes.

1. **Geometric distribution, variance (both sets).** The lecture notes write $\text{Var}(x) = P''(1) + P'(1) + \left[P'(1)\right]^2$; the sign of the last term should be minus. The extended set writes it correctly, and both sets carry the correct subsequent line $\frac{2q^2}{p^2} + \frac{q}{p} - \frac{q^2}{p^2}$, giving the correct answer q/p^2.
2. **Geometric distribution, $P''(1)$ (extended set).** The line reads $P''(1) = \frac{2pq^2}{(1-q)^3} = \frac{2q^2}{(1-q)^3} = \frac{2pq^2}{p^3}$; the middle expression is a slip — since $1-q=p$, it should be $\frac{2pq^2}{p^3} = \frac{2q^2}{p^2}$, which is the value used afterwards.
3. **Bernoulli distribution (extended set).** The line introducing $P(s) = q + ps$ says "The p.d.f. of x is given by"; it should read p.g.f.
4. **Bivariate variances (lecture notes).** The notes write $\sigma_x^2 = E[X(X-1)] + E(x)^2 - \left[E(x)\right]^2$; the first moment term should be $E(x)$, i.e. $\sigma_x^2 = E[X(X-1)] + E(x) - \left[E(x)\right]^2$ (compare §3.3, where $E(x^2) = E\{x(x-1)\} + Ex$ is used correctly). The companion formula for σ_y^2 is correct.
5. **Bivariate example (lecture notes).** The notes give $j, k = 0, 1, 2$; strictly X and Y range over all non-negative integers, the written range being the first few values.
6. **Random walk example (extended set).** The notes write "$\{X(n),\ t \in T\}$", using t where n is meant; and "moves one unit right to left" for "one unit to the right or to the left".
7. **Convergence interval (both sets).** The lecture notes give $s_0 < s < s_1$; the extended set gives $-s_0 < s < s_0$. Both are transcribed where they occur.

8. **Marginal notes.** Short margin notes ("Multiplying by s^n and summing over n", "Making $A(s)$ the subject", "Geometric series", "$P_x(s) = E(s^x)$") are reproduced in brackets at the points where they occur.

Appendix B. Coverage Map (source pages → sections)

Section	Extended set (7 pp.)	Lecture notes (15 pp.)
1. Stochastic processes: definition	p. 1	p. 1
1.2 Examples (4 in the extended set, 2 in the lecture notes)	p. 1 (i–iv)	p. 1 (i–ii)
2.1 Definition of the g.f.	pp. 1–2	p. 2
2.2 Coefficient extraction by differentiation	—	p. 2
2.3 Examples 1–3	p. 2	pp. 2–3
2.3 Example 4 (recurrence)	p. 2	pp. 3–4
3.1 p.g.f. definition	p. 3	p. 4
3.2 Sequence that is not a p.g.f.	p. 3	p. 4
3.3 Mean and variance from the p.g.f.	pp. 3–4	pp. 4–5
4.1 Bernoulli	p. 4	p. 5
4.2 Binomial	pp. 4–5	pp. 5–6
4.3 Geometric	p. 5	p. 6
4.4 Poisson	p. 6	p. 6 (marked "Read")
5.1 m.g.f. link	p. 6	p. 7
5.2 Integral of $P(s)$	p. 6	p. 7
5.3 p.g.f. of $Y = mX + n$	p. 6	p. 7
5.4 Recovering p_k; Poisson expansion	p. 7	pp. 7–8
6. Sums: fixed number	p. 7	pp. 8–9
6. Exp: sum of two Poissons	—	p. 9
7. Sums: random number (compound)	—	pp. 9–10
8.1 Upper tail	—	pp. 10–11
8.2 Lower tail	—	p. 12
9. Bivariate p.g.f., marginals, moments	—	pp. 13–14
9.3 Solved example	—	pp. 14–15

Every page of both sets is accounted for: extended set pages 1–7 and lecture notes pages 1–15.